

PROGRAM C: LINKED FILE ALLOCATION

C PROGRAM

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, blocks[20], i;
```

```
    printf("Enter Number of Blocks: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter Block Numbers:\n");
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        scanf("%d", &blocks[i]);
```

```
    }
```

```
    printf("\nLinked Allocation:\n");
```

```
    for (i = 0; i < n - 1; i++)
```

```
    {
```

```
        printf("%d --> ", blocks[i]);
```

```
    }
```

```
    printf("%d --> NULL\n", blocks[n - 1]);
```

```
    return 0;
```

}

SHELL SCRIPT

```
GNU nano 8.7.1
#!/bin/bash

echo "Enter Number of Blocks:"
read n

echo "Enter Block Numbers:"

for ((i = 0; i < n; i++))
do
    read block[$i]
done

echo "Linked Allocation:"

for ((i = 0; i < n - 1; i++))
do
    echo -n "${block[$i]} --> "
done
```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano linked.sh
(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x linked.sh
(ganesh@LAPTOP-R3RDH9NG)~$ bash linked.sh
Enter Number of Blocks:
5
Enter Block Numbers:
4
3
2
1
6
Linked Allocation:
4 --> 3 --> 2 --> 1 -->
```